

FraudGuard AI

Team Name: Code Hackerz

Project Title: FraudGuard AI – Real-Time Fraud Detection System

Hackathon: Samsung PRISM GenAI Hackathon 2025

Theme: AI for Core Applications

Institution: SRM Institute of Science & Technology, Kattankulathur

Team Members

Name	Role
Arushi Jain	AI Model & Research
Jasmitha Lokesh	Backend & Integration
Gnana PrasunaM	Frontend & Dashboard

Problem Statement

Financial fraud is growing rapidly due to sophisticated online scams.

Traditional rule-based systems are slow, reactive, and easy to bypass.

We need an AI-powered, real-time fraud detection system that flags suspicious transactions **instantly**.

Objective

Build a **real-time fraud detection platform** that:

- Analyzes transactions instantly.
- Flags high-risk activities using ML-based scoring.
- Provides an interactive dashboard for monitoring.

System Architecture

Workflow:

1. **Data Input:** Transaction details (amount, device, location, time).
 2. **Feature Extraction:** Behavior patterns, risk scoring.
 3. **AI Engine:** ML model evaluates fraud risk.
 4. **Backend API:** FastAPI provides prediction endpoints.
 5. **Frontend Dashboard:** Visualizes transaction stream & alerts.
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Tech Stack

Layer	Tools/Frameworks
Frontend	HTML, Tailwind CSS, JavaScript
Backend	Python (FastAPI)
ML	Scikit-learn, NumPy, Pandas
Visualization	Tailwind UI Dashboard
Others	Joblib, Uvicorn

Key Features

Real-time monitoring of simulated transactions
Risk-based color alerts (Low, Medium, High)
AI-driven fraud probability scoring
Intuitive dashboard for insights

Impact

- Detects fraudulent activity in **real-time**
 - Reduces false positives
 - Improves customer trust
 - Can be scaled across **banks, fintechs, and payment systems**
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Submissions

- **Presentation Slides:**
<https://docs.google.com/presentation/d/1yWhewirChc03nfGZyLYGWtA6xTNMs-y/edit?slide=id.p1#slide=id.p1>
 - **GitHub Repo:** <https://github.com/dare-devil-96/Real-time-fraud-detection>
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Summary

FraudGuard AI provides a **real-time, AI-driven** approach to prevent online frauds before they occur.

Built by **Code Hackerz**, it's lightweight, scalable, and ready for integration into real-world fintech systems.